

ASSIGNMENT # 02

Subject: PF



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Question #01

Part a

Code:

```
#include<stdio.h>

//part a

//function main begins program execution

int main(void)
{
    unsigned int x;
    unsigned int y;
    unsigned int i;
    unsigned int j;
    //prompt user for input
    printf("%s", "Enter two unsigned integers in the range 1-20 :");
    scanf("%u%u", &x , &y ); //read values for x and y

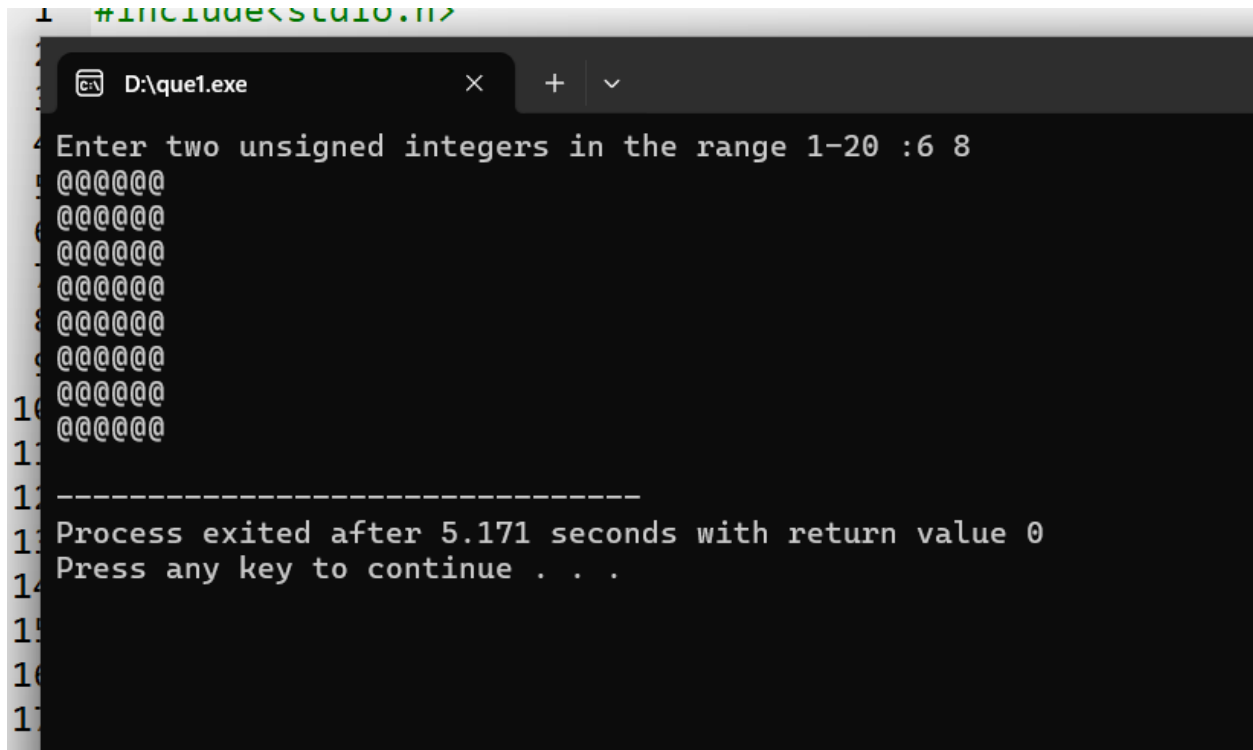
    for ( i=1; i<=y; ++i) { //count from 1 to y

        for( j=1; j<=x; ++j) { //count from 1 to x
            printf("%s" , "@"); // output@
        } // end inner for

        puts( "" ); //begin new line
```

```
    } // end outer for  
} // end function main
```

screenshot:



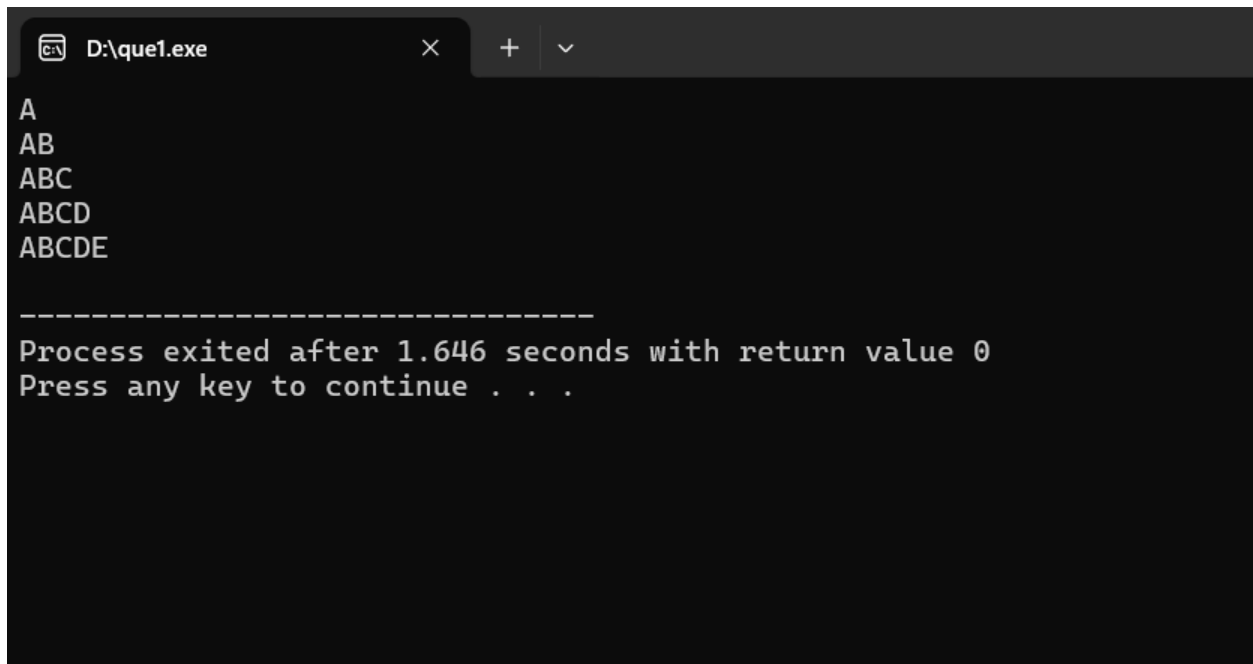
Part b

Code:

```
#include<stdio.h>  
  
int main(){  
    //part b  
    for(char ch='A'; ch<='E'; ++ch){  
        for(char c='A'; c<=ch; ++c){  
            printf("%c,",c);  
        }  
        printf("\n");  
    }
```

```
    }  
    return 0;  
}
```

Screenshot:



```
D:\que1.exe  
A  
AB  
ABC  
ABCD  
ABCDE  
  
-----  
Process exited after 1.646 seconds with return value 0  
Press any key to continue . . .
```

Question #02

Code:

```
#include<stdio.h>  
  
int main(){  
    //part a  
    for (int i=1; i<=7; i++) {  
        printf("%d, ", i);  
    }  
    printf("\n");  
    //part b  
    for (int i=3; i<=23; i+=5) {
```

```

        printf("%d, ", i);
    }
    printf("\n");
//part c
    for (int i=20; i>=-10; i-=6) {

        printf("%d, ", i);
    }
    printf("\n");
//part d
    for (int i=19; i<=51; i+=8) {

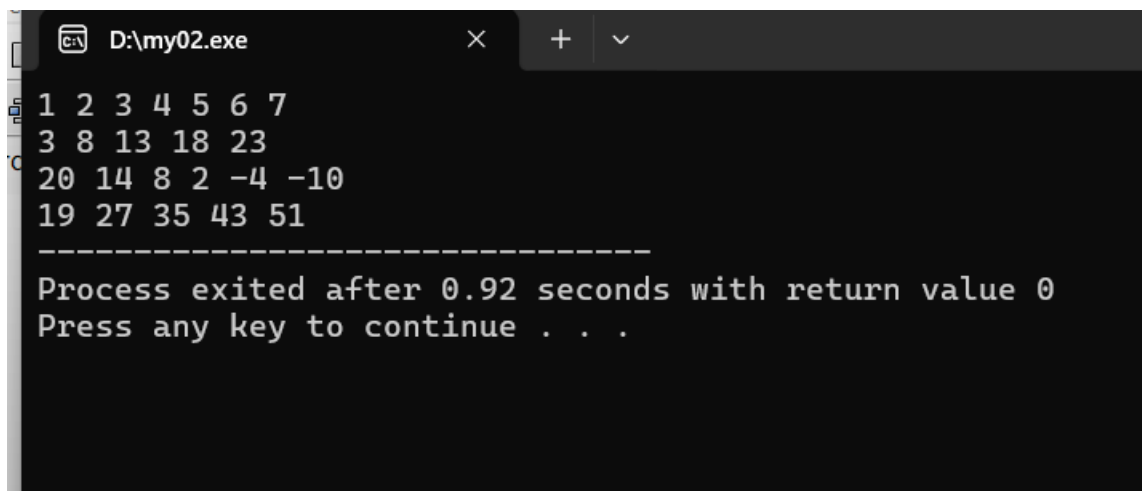
        printf("%d, ",i);
    }

}

}

```

ScreenShot:



```

D:\my02.exe
1 2 3 4 5 6 7
3 8 13 18 23
20 14 8 2 -4 -10
19 27 35 43 51
-----
Process exited after 0.92 seconds with return value 0
Press any key to continue . . .

```

Question #03

Part a:

Pseudocode:

start

1. Prompt the user for the value of x .
2. Prompt the user for the total number of terms n .
3. Initialize the sum to 1 (the first term of the series).
4. Initialize a variable `term` to 1 (this will be the current term in the series).
5. Loop from 1 to $n-1$ (because the first term is already added):
 - Update the term: $\text{term} = (\text{term} * 2) / x$.
 - Add the term to the sum.
6. Print the sum.

end

code:

```
#include <stdio.h>
```

```
int main() {
```

```
    //part a
```

```
    int n;
```

```
    double x, sum = 0.0, term = 1.0;
```

```
    // Prompt user for the number of terms
```

```
    printf("Enter the number of terms: ");
```

```
    scanf("%d", &n);
```

```
    // Prompt user for the value of x
```

```
    printf("Enter the value of x: ");
```

```
    scanf("%lf", &x);
```

```
    // Loop to calculate the sum of the series
```

```
    for (int i = 0; i < n; i++) {
```

```

    sum += term;

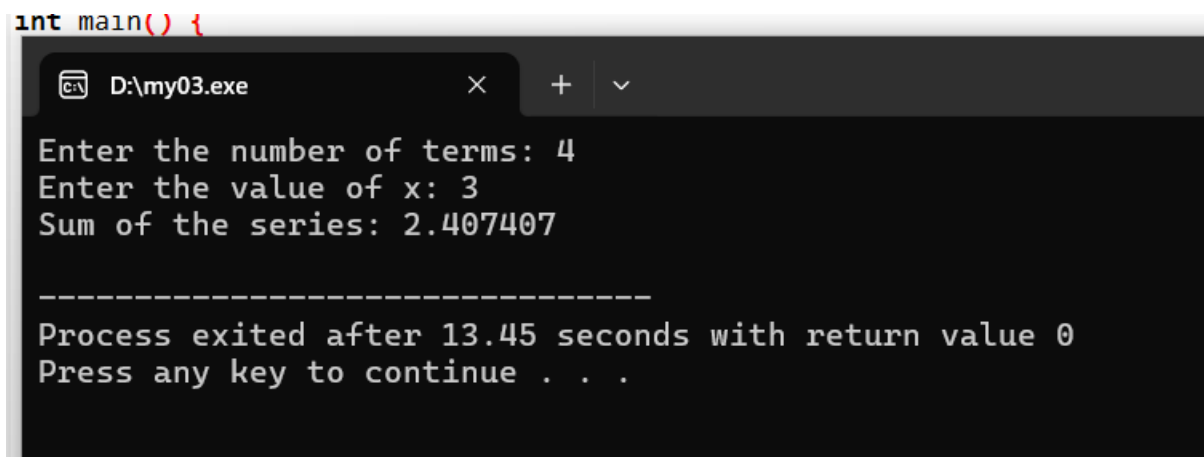
    term = (term * 2) / x;
}

// Print the sum of the series
printf("Sum of the series: %lf\n", sum);

return 0;
}

```

ScreenShot:



```

int main() {
    Enter the number of terms: 4
    Enter the value of x: 3
    Sum of the series: 2.407407

    -----
    Process exited after 13.45 seconds with return value 0
    Press any key to continue . . .

```

Part b:

Pseudocode:

start

1. Prompt the user for the value of x .
2. Prompt the user for the total number of terms n .
3. Initialize the sum to 2 (the first term of the series).
4. Initialize a variable `term` to 5 (this will be the current term in the series).
5. Initialize a variable `power_of_x` to 1 (this will hold successive powers of x).
6. Loop from 1 to $n-1$ (because the first term is already added):
 - o Update the term: `term = term * 5 / x`.
 - o Update `power_of_x`: `power_of_x = power_of_x * x`.

- Add `term` to the sum.
7. Print the sum.

end

Code:

```
#include <stdio.h>
```

```
int main() {
```

```
    //part b
```

```
    int n;
```

```
    double x, sum = 2, term = 5;
```

```
    // Prompt the user for the value of x and the number of terms
```

```
    printf("Enter the value of x: ");
```

```
    scanf("%lf", &x);
```

```
    printf("Enter the total number of terms: ");
```

```
    scanf("%d", &n);
```

```
    // Loop to calculate the sum of the series
```

```
    for (int i=1; i < n; i++) {
```

```
        sum += term;
```

```
        term = term*(5/x);
```

```
    }
```

```
    // Print the result
```

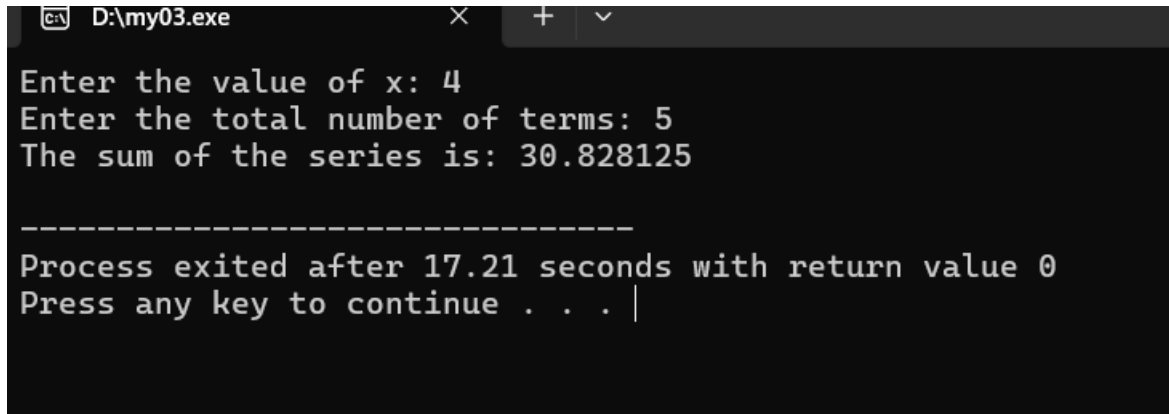
```
    printf("The sum of the series is: %lf\n", sum);
```

```
    return 0;
```



```
}
```

Screenshot:



```
D:\my03.exe
Enter the value of x: 4
Enter the total number of terms: 5
The sum of the series is: 30.828125

-----
Process exited after 17.21 seconds with return value 0
Press any key to continue . . . |
```

Question #04

Code:

```
#include<stdio.h>

int main(){
    int distance;
    int speed;

    printf("enter speed of vehicle(in miles per hour) :");
    scanf("%d",&speed);

    int hours;

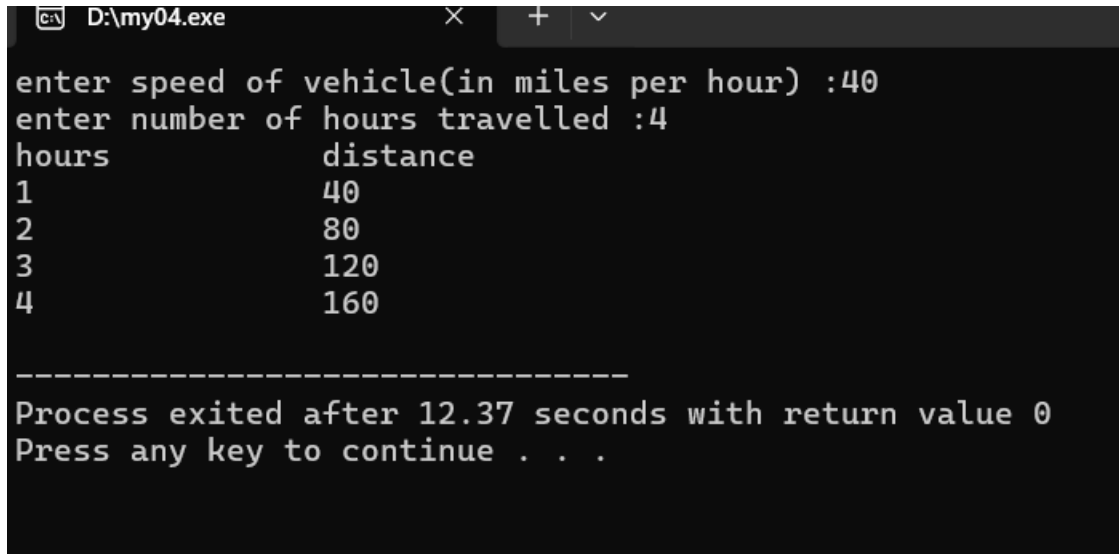
    printf("enter number of hours travelled :");
    scanf("%d",&hours);

    printf("hours\t\tdistance\n");

    for(int i=1; i<=hours; i++)    {
        distance=speed*i;
        printf("%d\t\t%d\n",i,distance);
    }
}
```

```
}
```

Screenshot:



```
D:\my04.exe
enter speed of vehicle(in miles per hour) :40
enter number of hours travelled :4
hours      distance
1          40
2          80
3          120
4          160

-----
Process exited after 12.37 seconds with return value 0
Press any key to continue . . .
```

Question #05

Pseudocode:

start

1. Initialize counters for each category and total values for each category.
2. Loop to read product number and average quantity sold pairs.
3. Use switch statement to determine the retail price based on product number.
4. Update the counters and total values based on product category.
5. Print the number of products sold for each category.
6. Print the total retail value for each category.
7. Print the total retail value of all products sold.

end

Code:

```
#include <stdio.h>
```

```
int main() {
```

```
    int product_number, quantity;
```

```
    int count_low = 0, count_medium = 0, count_high = 0;
```

```
double total_low = 0, total_medium = 0, total_high = 0;

double retail_price;

double total_retail_value = 0;


printf("Enter product number and average quantity sold for one day (end input with -1 -
1):\n");

while (1) {

    scanf("%d %d", &product_number, &quantity);

    if (product_number == -1 && quantity == -1) {

        break;

    }


    switch (product_number) {

        case 1:

            retail_price = 200;

            count_low += quantity;

            total_low += retail_price * quantity;

            break;

        case 2:

            retail_price = 250;

            count_low += quantity;

            total_low += retail_price * quantity;

            break;

        case 3:

            retail_price = 530;

            count_medium += quantity;

            total_medium += retail_price * quantity;
```

```

        break;
    case 4:
        retail_price = 600;
        count_medium += quantity;
        total_medium += retail_price * quantity;
        break;
    case 5:
        retail_price = 650;
        count_medium += quantity;
        total_medium += retail_price * quantity;
        break;
    case 6:
        retail_price = 1100;
        count_high += quantity;
        total_high += retail_price * quantity;
        break;
    default:
        printf("Invalid product number.\n");
        continue;
}

total_retail_value += retail_price * quantity;
}

printf("\nProducts Sold Last Week:\n");
printf("Low category: %d products\n", count_low);
printf("Medium category: %d products\n", count_medium);

```

```

printf("High category: %d products\n", count_high);

printf("\nTotal Retail Value:\n");

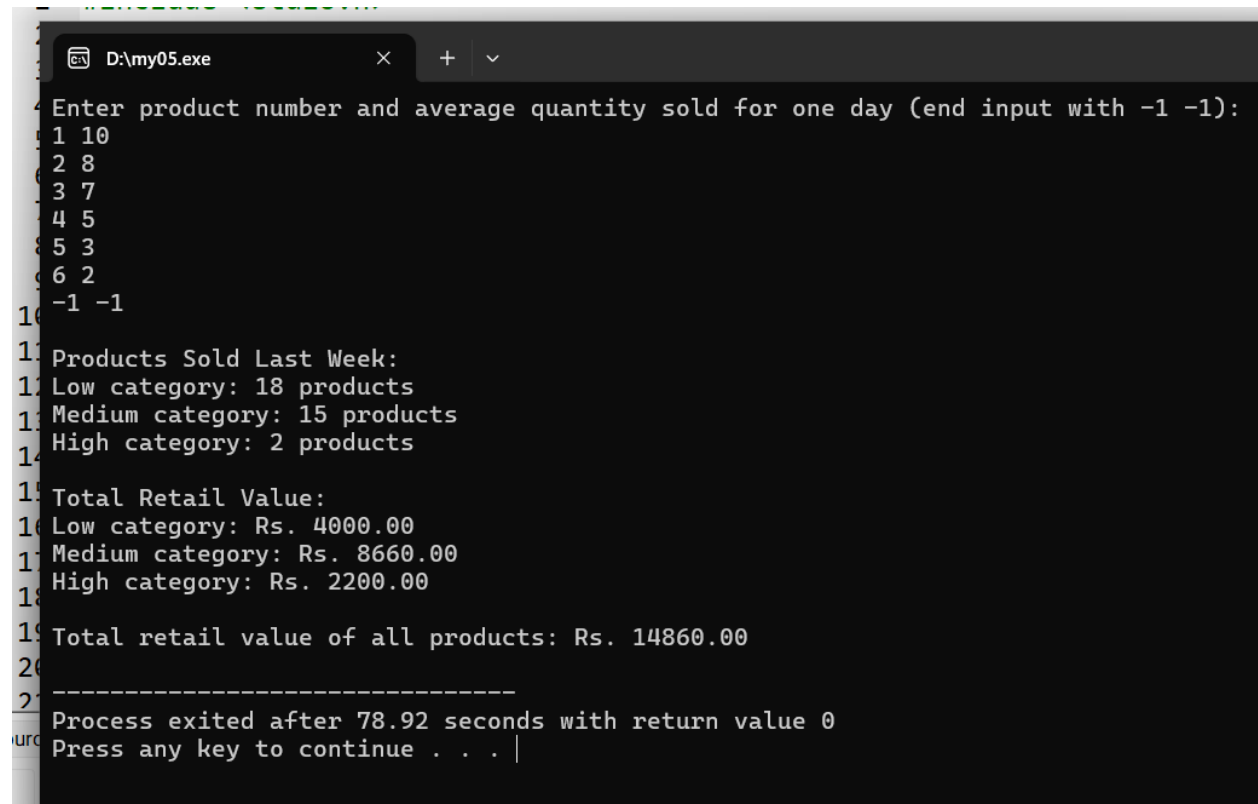
printf("Low category: Rs. %.2f\n", total_low);
printf("Medium category: Rs. %.2f\n", total_medium);
printf("High category: Rs. %.2f\n", total_high);

printf("\nTotal retail value of all products: Rs. %.2f\n", total_retail_value);

return 0;
}

```

ScreenShot:



```

D:\my05.exe
Enter product number and average quantity sold for one day (end input with -1 -1):
1 10
2 8
3 7
4 5
5 3
6 2
-1 -1
Products Sold Last Week:
Low category: 18 products
Medium category: 15 products
High category: 2 products
Total Retail Value:
Low category: Rs. 4000.00
Medium category: Rs. 8660.00
High category: Rs. 2200.00
Total retail value of all products: Rs. 14860.00
-----
Process exited after 78.92 seconds with return value 0
Press any key to continue . . .

```

Question #06

Code:

```
#include <stdio.h>
```

```
int main() {
```

```
    int paycode;
```

```
    double weekly_pay;
```

```
    double weekly_salary;
```

```
    double hourly_wage;
```

```
    int hours_worked;
```

```
    double gross_sales;
```

```
    double fixed_amount_per_item;
```

```
    int items_produced;
```

```
    printf("Enter paycode for employee (1 for Manager, 2 for Hourly Worker, 3 for Commission Worker, 4 for Pieceworker, -1 to end):\n");
```

```
    while (1) {
```

```
        printf("Paycode: ");
```

```
        scanf("%d", &paycode);
```

```
        if (paycode == -1) {
```

```
            break;
```

```
        }
```

```
        switch (paycode) {
```

```
            case 1: // Manager
```

```
printf("Enter weekly salary: ");  
scanf("%lf", &weekly_salary);  
weekly_pay = weekly_salary;  
printf("Manager's pay: Rs. %.2f\n", weekly_pay);  
break;
```

case 2: // Hourly worker

```
printf("Enter hourly wage: ");  
scanf("%lf", &hourly_wage);  
printf("Enter hours worked: ");  
scanf("%d", &hours_worked);  
if (hours_worked > 40) {  
    weekly_pay = (40 * hourly_wage) + ((hours_worked - 40) * hourly_wage * 1.5);  
} else {  
    weekly_pay = hours_worked * hourly_wage;  
}  
printf("Hourly worker's pay: Rs. %.2f\n", weekly_pay);  
break;
```

case 3: // Commission worker

```
printf("Enter gross weekly sales: ");  
scanf("%lf", &gross_sales);  
weekly_pay = 250 + (0.057 * gross_sales);  
printf("Commission worker's pay: Rs. %.2f\n", weekly_pay);  
break;
```

case 4: // Pieceworker

```
printf("Enter fixed amount per item: ");  
scanf("%lf", &fixed_amount_per_item);  
printf("Enter number of items produced: ");
```

```

        scanf("%d", &items_produced);

        weekly_pay = fixed_amount_per_item * items_produced;

        printf("Piecworker's pay: Rs. %.2f\n", weekly_pay);

        break;
default:

        printf("Invalid paycode.\n");

        break;

    }

}

return 0;

}

```

ScreenShot:

```

D:\my06.exe
Enter paycode for employee (1 for Manager, 2 for Hourly Worker, 3 for Commission Worker, 4 for Piecworker, -1 to end):
Paycode: 1
Enter weekly salary: 1200
Manager's pay: Rs. 1200.00
Paycode: 2
Enter hourly wage: 20
Enter hours worked: 42
Hourly worker's pay: Rs. 860.00
Paycode: 3
Enter gross weekly sales: 7000
Commission worker's pay: Rs. 649.00
Paycode: 4
Enter fixed amount per item: 15
Enter number of items produced: 200
Piecworker's pay: Rs. 3000.00
Paycode: -1

-----
Process exited after 132.9 seconds with return value 0
Press any key to continue . . .

```

Question #07

Code:


```
#include <stdio.h>
```

```
int main() {
```

```
    int rows = 10; // Number of rows for the triangle
```

```
    // Pattern 1: Right-angled triangle
```

```
    printf("Pattern 1: Right-angled triangle\n");
```

```
    for (int i = 1; i <= rows; i++) {
```

```
        for (int j = 1; j <= i; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    printf("\n");
```

```
    // Pattern 2: Inverted right-angled triangle
```

```
    printf("Pattern 2: Inverted right-angled triangle\n");
```

```
    for (int i = rows; i >= 1; i--) {
```

```
        for (int j = 1; j <= i; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    printf("\n");
```

```
    // Pattern 3: Left-aligned right-angled triangle
```

```
    printf("Pattern 3: Left-aligned right-angled triangle\n");
```

```
for (int i = 1; i <= rows; i++) {  
    for (int j = 1; j <= rows; j++) {  
        if (j <= rows - i) {  
            printf(" ");  
        } else {  
            printf("*");  
        }  
    }  
    printf("\n");  
}  
printf("\n");
```

// Pattern 4: Centered right-angled triangle

```
printf("Pattern 4: Centered right-angled triangle\n");  
for (int i = 1; i <= rows; i++) {  
    for (int j = 1; j <= rows - i; j++) {  
        printf(" ");  
    }  
    for (int k = 1; k <= i; k++) {  
        printf("*");  
    }  
    printf("\n");  
}
```

```
return 0;
```

```
}
```

ScreenShot:

Pattern 1: Right-angled triangle

```
*  
**  
***  
****  
*****  
*****  
*****  
*****  
*****  
*****
```

Pattern 2: Inverted right-angled triangle

```
*****  
*****  
*****  
*****  
*****  
*****  
*****  
***  
**  
*
```

Pattern 3: Left-aligned right-angled triangle

```
 *
**
***
****
*****
*****
*****
*****
*****
*****
```

Pattern 4: Centered right-angled triangle

```
 *
**
***
****
*****
*****
*****
*****
*****
*****
```

Process exited after 0.7111 seconds with return value 0
Press any key to continue . . .

Question #08

Code:

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
float pH;
```

```
printf("enter the pH value :");
```

```
scanf("%f",&pH);
```

```
if(pH>7){
```

```
    if(pH<12){
        printf("the solution is alkaline\n");
    }
    else{
        printf("the solution is very alkaline\n");
    }

} else{
    if(pH==7){
        printf("the solution is neutral\n");
    } else{
        if(pH>2){
            printf("the solution is Acidic\n");
        }
        else{
            printf("the solution is very Acidic \n");
        }
    }
}
}
```

ScreenShot:

Command line

```
D:\>my08
enter the pH value :7
the solution is neutral

D:\>gcc -o my08 my08.c

D:\>my08
enter the pH value :6
the solution is Acidic

D:\>gcc -o my08 my08.c

D:\>my08
enter the pH value :1
the solution is very Acidic

D:\>gcc -o my08 my08.c

D:\>my08
enter the pH value :9
the solution is alkaline

D:\>gcc -o my08 my08.c

D:\>my08
enter the pH value :13
the solution is very alkaline

D:\>
```

